

INTERNSHIP REPORT

FIBER OPTIC COMMUNICATIONS, FTTH NETWORK ARCHITECTURE AND HOME APPLIANCES

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CERTIFICATE

This is to certify that the internship report entitled "Fiber Optic Communications, FTTH Network Architecture and Electrical Home Appliances Maintenance" is a bona fide record of the industrial training carried out by ANWAR PC under my academic supervision and guidance, in corporate collaboration with Bfone Fibernet, Edavannappara, in partial fulfillment of the requirements for the award of the Degree of Bachelor of Science in Physics.

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INTERNSHIP ACTIVITY SCHEDULE

Day	Date	Topics / Activities Covered
Day 1	21.05.2026	Optical Fiber Theory Class (Introduction to OFC)
Day 2	22.05.2026	Splicing Practical Section
Day 3	23.05.2026	Repairing of the Ceiling Fan
Day 4	25.05.2026	• Theory Class of Fiber To The Home (FTTH) • Repairing of Mixer Grinder • Field Visit of Splicing
Day 5	26.05.2026	Field Visit at Manjeri Telephone Exchange
Day 6	08.06.2026	Power Calculation Theory Class
Day 7	09.06.2026	• Repairing of Iron Box • Repairing of Wall Fan • Repairing of Induction Cooker

Table 1: Day-wise Internship Activity Log

Part I

Fiber Optic Communications and FTTH Network Architecture

PART INTRODUCTION: FIBER OPTIC INFRASTRUCTURE

Modern communications are fundamentally driven by high-speed data transfer infrastructures operating on optical physics principles. Transitioning away from traditional legacy copper wire infrastructures, optical fibers utilize light pulses to transmit vast amounts of data over immense distances with minimal attenuation. This part of the internship report establishes a detailed technical study of modern Fiber-Optic Communication (OFC) and Fiber-to-the-Home (FTTH) network architectures, directly informed by industrial training with Bfome Fibernet. The core purpose here is to outline the entire journey of the light signal, from the hardware source down to the home ONT. The chapters herein analyze the physical principles of Total Internal Reflection (TIR) that guide light within glass matrices, the structural mechanics of modern multi-core cables, and the quantitative engineering required for link budgeting and structural calculations. Moreover, we document standard field procedures, including precision splicing, color code identification, and troubleshooting techniques used by technicians. This section finishes with a technical summary of our field study at the Manjeri BSNL Exchange Office.

CHAPTER 1

INTRODUCTION TO OPTICAL FIBER COMMUNICATION

1.1 Overview

Optical Fiber Communication (OFC) refers to the technique of transferring information across distances using modulated light pulses guided through thin fibers. The light forms an electromagnetic carrier wave that is modulated to carry information.

1.2 The Basic Fiber Optic Link

A standard fiber-optic link is made up of a sequence of core hardware blocks:

- Encoder/Source: Converts data into electrical signals.
- Transmitter (E/O): Converts electrical signals into optical signals using LEDs or LASERs.
- Optical Fiber: The medium that guides the light signal.
- Receiver (O/E): Uses a photo-detector (PIN or APD) to convert light back into electrical signals.
- Decoder: Recovers the original data from the signal.

1.3 Advantages of Fiber Optics

- Non-conductive: No risk of sparks or electrical hazards.
- Electromagnetic Immunity: Immune to EMI and lightning.
- Large Bandwidth: Can carry massive amounts of data.

- Low Loss: Signal attenuation is significantly lower than copper cables.
- Small and Lightweight: Easier to install and transport.

CHAPTER 2

PRINCIPLE AND CABLE STRUCTURE

2.1 Principle of Operation: Total Internal Reflection

Light is guided along the length of the optical fiber core using the principle of Total Internal Reflection (TIR). For TIR to occur, two conditions must be satisfied:

- The refractive index of the core (n_1) must be greater than the cladding (n_2).
- The angle of incidence must be greater than the critical angle (θ_c).

The critical angle value is determined by:

$$\theta_c = \sin^{-1}(n_2 / n_1) \quad (2.1)$$

2.2 Cable Anatomy

The physical architecture of a standard fiber cable consists of these core elements:

- Core: The innermost glass or plastic part where light travels.
- Cladding: Layer with a lower refractive index that reflects light back into the core.
- Buffer: Protects the fiber from moisture.
- Strength Member: Usually Aramid yarn (Kevlar) to prevent cable breakage.
- Outer Jacket: Protects against environmental damage.

2.3 Cable Types

- Tight-Buffer Cable: Used mainly for indoor applications and patch cords.
- Loose-Tube Cable: Used for outdoor environments (ducts, direct burial).
- Ribbon Cable: Dozens of fibers packed together for high-density applications.

CHAPTER 3

LINK BUDGETING, SPLICING, AND MATHEMATICAL CALCULATIONS

3.1 OFC Splicing

Splicing is used to make permanent connections between fiber ends. The two main types are:

- Fusion Splicing: Using an electric arc to melt and fuse fibers.
- Mechanical Splicing: Aligning fibers using a mechanical fixture.

3.2 Logarithmic Conversion Mechanics

To make loss calculations additive rather than multiplicative, optical power is measured using the logarithmic decibel-milliwatt (dBm) scale:

$$P_{\text{dBm}} = 10 \log_{10}(P_{\text{mW}} / 1 \text{ mW}) \quad (3.1)$$

3.3 Splitter Loss Calculations

An optical splitter's theoretical insertion loss increases mathematically with the number of output ports (N):

$$\text{Loss}_{\text{Ideal}} = 10 \log_{10}(N) \quad (3.2)$$

3.4 Standard Loss Values for Link Performance

For network design and planning, the standard component attenuation values are defined as:

Network Element Component	Standard Loss Value
Fiber Path Loss @ 1310 nm / 1490 nm	0.40 dB per km
Fiber Path Loss @ 1550 nm	0.25 dB per km
Fusion Splice Junction	0.10 dB per splice
SC/UPC Physical Connector Pair	0.35 dB per connector
SC/APC Physical Connector Pair	0.25 dB per connector
1 × 2 Splitter Module	3.0 dB average insertion loss
1 × 16 Splitter Module	13.0 dB average insertion loss
1 × 32 Splitter Module	16.0 dB average insertion loss

Table 3.1: Standard Loss Parameters Used in FTTH Networks

3.5 Mathematical Verification: Power Link Budget Case Study

$$P_r = P_t - \text{Loss}_{\text{total}} \quad (3.3)$$

3.5.1 Case Study Parameter Blueprint

As a practical case study, consider a 7 km fiber link installed by Bfone Fibernet. The link features 4 fusion junctions, 2 SC/UPC connector pairs, and a cascaded splitter configuration consisting of a 1 × 16 and a 1 × 8 unit. The total link budget metrics break down into the following distinct attenuation values:

- Fiber Line Attenuation Loss: $7 \text{ km} \times 0.40 \text{ dB/km} = 2.8 \text{ dB}$
- First Stage Insertion Loss (1 × 16 Splitter) = 13.0 dB
- Second Stage Insertion Loss (1 × 8 Splitter) = 10.0 dB
- Total Mechanical Splice Losses: $4 \times 0.10 \text{ dB} = 0.4 \text{ dB}$
- Total Dynamic Connector Losses: $2 \times 0.35 \text{ dB} = 0.7 \text{ dB}$

The sum of these individual components represents the total link attenuation:

$$\begin{aligned} \text{Total Attenuation Loss} &= 2.8 + 13.0 + 10.0 + 0.4 + 0.7 \\ &= 26.9 \text{ dB} \end{aligned}$$

3.5.2 Link Status Analysis

The acceptable operating window for standard ONT fiber modems is between -10 dBm and -28 dBm: Receiver Sensitivity Threshold $\leq -28 \text{ dBm}$

$$\text{Receiver Sensitivity Threshold} \leq -28 \text{ dBm} \quad (3.4)$$

With a calculated link loss of 26.9 dB and a transmitter power of 0 dBm, the signal strength at the customer premises is -26.9 dBm. Because this value is stronger than the -28 dBm sensitivity limit, the link will operate successfully.

CHAPTER 4

OPTICAL FIBER SPLICING: DETAILED ANALYSIS

4.1 What is Splicing?

Splicing refers to the permanent joining of two optical fiber cores to allow continuous light propagation. Unlike connectors, which are designed for frequent re-mating, splices are meant to be enduring and stable joints. The primary goal is to minimize optical loss and prevent back-reflection.

4.2 Types of Splicing

Technicians utilize two primary methods for joining fibers:

4.2.1 Fusion Splicing

This is the premium industry-standard method, which utilizes an electric arc to melt the glass tips and fuse them permanently. Loss: Typically 0.01 to 0.05 dB. Usage: Backbone networks, long-haul links, and FTTH.

4.2.2 Mechanical Splicing

This technique uses a mechanical alignment sleeve containing index-matching gel to hold the fibers in alignment. Loss: Typically 0.2 to 0.5 dB. Usage: Emergency repairs, temporary setups, and environments where fusion equipment is not available.

4.3 Importance and Significance

- Link Extension: Connecting multiple segments to build long-distance networks.
- Signal Integrity: Maintaining near-perfect light transmission with minimal attenuation.
- Back-reflection Control: Minimizing return loss, which protects laser sources.
- Durability: Providing maintenance-free connections that withstand environmental stress.

4.4 Step-by-Step Fusion Splicing Process

1. Stripping: The 250 μm outer coating is removed using specialized fiber strippers without scratching the glass.
2. Cleaning: The bare glass fiber is thoroughly cleaned using 99% Isopropyl Alcohol (IPA) and lint-free wipes to remove all contaminants.
3. Cleaving: The fiber is cut using a precision diamond-blade cleaver to ensure a perfect 90-degree end face (angle $< 0.5^\circ$).
4. Fusion: The fiber ends are aligned using the Profile Alignment System (PAS) and fused using an electric arc at 2000°C.
5. Protection: A heat-shrink splice sleeve is applied and heated to protect the joint from physical damage and moisture.

4.5 Advantages

- Lowest possible insertion loss.
- Permanent and highly reliable joints.
- Resistant to vibration and temperature fluctuations.
- Space-efficient for high-density fiber enclosures.

4.6 Future Scope

- Automation: Developing fully autonomous AI-driven splicing robots for sub-sea and hazardous environments.
- Multi-core Fiber Splicing: Moving beyond single-core to support high-density multi-core fibers to increase bandwidth capacity.
- Enhanced Precision: Integrating advanced nanotechnology for near-zero loss splicing in extreme environments.

4.7 Conclusion

Ultimately, fiber splicing is a core skill in optical telecommunications. Permanent joins minimize attenuation, allowing light signals to traverse long distances. For primary network segments, fusion splicing is dominant, whereas mechanical splicing remains useful for emergency restoration work.

CHAPTER 5

INTRODUCTION TO FTTH & PASSIVE OPTICAL NETWORKS (PON)

5.1 Overview of FTTH

FTTH represents the transition to complete optical networking, replacing copper phone networks (DSL) and coax networks with dedicated optical fibers directly to the home. Most modern FTTH networks utilize PON (Passive Optical Network) technology. The term “passive” is critical: it means the outer distribution network requires no electrical power between the provider’s central exchange and the customer’s home. Signals are routed and split using optical principles alone, drastically lowering operational costs, minimizing equipment failure due to power surges, and maximizing lifespan. This point-to-multipoint architecture allows a single optical fiber to serve multiple premises through splitters.

5.2 The FTTH Network Architecture Layers

An FTTH network's physical layout is categorized into three key segments:

- A. The Feeder Segment: Starts at the service provider’s Central Office (CO) and runs to a local neighborhood hub known as the Fiber Distribution Hub (FDH) or Primary Cross-Connect box. This segment uses high-core-count cables (often containing 48 to 144+ fibers) designed to carry aggregated data over long distances (up to 20 kilometers) without splitting.
- B. The Distribution Segment: Travels from the FDH to smaller street-level ODP (Optical Distribution Point) or FAT (Fiber Access Terminal) boxes. At the FDH, the feeder fibers are connected to splitters, and the split fibers branch out into the neighborhood using medium-core cables (12 to 24 fibers) along utility poles or underground conduits.
- C. The Drop Segment: The final link that bridges the ODP/FAT on the street directly into the customer’s living room or office. This utilizes a Drop Cable, which is a highly specialized, lightweight, ruggedized fiber cable (usually containing only 1 or 2 fibers) engineered to survive tight bends and physical pulling.

5.3 Core Hardware Components

FTTH operations depend on a clear split between active and passive network elements:

5.3.1 Active Components (Power Required)

OLT (Optical Line Terminal): Installed at the Central Office (CO). It acts as the central 'brain' of the network, converting digital IP data from the backbone into laser pulses and scheduling traffic for thousands of subscribers.

ONT / ONU (Optical Network Terminal / Unit): Installed inside the subscriber's home. It acts as the translator, receiving optical light signals and converting them back into standard electrical signals (Ethernet/Wi-Fi) for consumer devices.

5.3.2 Passive Components (No Power Required)

Optical Splitters: The heart of PON. A passive glass prism component that takes a single incoming beam of light and splits it into multiple identical streams to serve multiple users. Standard split ratios are 1:8, 1:16, 1:32, or 1:64.

ODF / FDH Hubs: Environmental enclosures or cabinets that house splices and splitters, protecting connections from water, dust, and temperature changes.

CHAPTER 6

FTTH DEPLOYMENT, CODING AND TESTING

6.1 Fiber Color Coding Standard (TIA-598-C)

To manage multi-core cables without confusion, a standard 12-color sequence is followed. When core counts exceed 12, the color code is repeated with a colored tracer thread.

No.	Fiber Color	Symbol	No.	Fiber Color	Symbol
1	Blue	BL	7	Red	RD
2	Orange	OR	8	Black	BK
3	Green	GR	9	Yellow	YL
4	Brown	BR	10	Violet	VI
5	Slate (Grey)	SL	11	Rose (Pink)	RS
6	White	WH	12	Aqua	AQ

Table 6.1: TIA-598-C Color Coding Identification

6.2 Physical Deployment & Site Survey

6.2.1 Site Survey & Planning

The deployment process begins with a detailed site survey comprising:

- Route Mapping: Finding the shortest, most efficient path from street FAT to house.
- Structural Assessment: Identifying building entry points and optimal indoor ONT placement.
- Safety Check: Keeping clear of electrical mainlines to avoid physical interference.
- BOM Calculation: Generating the exact Bill of Materials (cables, clamps, rosettes).

6.2.2 Outdoor Routing Operations

Aerial Deployment: Fiber cables are strung between utility poles using tension/anchor clamps. Slack loops must be left at ends to accommodate weather-induced physical shifts. Underground Deployment: In suburban areas, fiber is pulled through buried protective HDPE ducts to shield it from excavation and weather damage.

6.2.3 Indoor Termination

The drop cable passes into the home and terminates inside an Access Terminal Box (ATB) or Fiber Rosette on the wall. A bend-insensitive patch cord (G.657 standard) is then run from the outlet to the ONT, allowing tight bends around walls without causing signal loss.

6.3 Splicing Methods Comparison

Feature	Fusion Splicing	Mechanical / Fast Connectors
Mechanism	Uses a high-voltage electric arc to melt and fuse glass tips permanently.	Physically aligns two fiber ends inside a sleeve using index-matching gel.
Signal Loss	Extremely low (< 0.05 dB to 0.1 dB).	Slightly higher (0.2 dB to 0.5 dB).
Pros & Cons	Most reliable, but requires an expensive fusion splicer machine.	Fast setup, low initial tool cost. Ideal for home drop termination.

Table 6.2: Comparison of Fusion and Mechanical Splicing

6.4 Link Budget & Testing Benchmarks

To prevent collisions on the single-core fiber, different wavelengths are used for upload and download:

6.4.1 Wavelength Multiplexing

- 1490 nm: Downstream data/voice traffic (OLT to ONT).
- 1310 nm: Upstream data upload traffic (ONT to OLT).
- 1550 nm: Optional downstream video/TV overlay.

6.4.2 Testing Parameters

- Optical Power Meter (OPM): Measures absolute light strength at the subscriber end. Acceptable operational range is between -15 dBm and -25 dBm. Readings below -27 dBm constitute an under-power fault.
- OTDR: Measures backscattered light to map the entire physical link and locate precise coordinates of cuts or defects.

6.5 Troubleshooting & Indicators

ONT faceplate status LEDs provide primary diagnostic information during service calls:

ONT Indicator	Status State	Meaning / Corrective Action
LOS (Red)	Steady Glow	Loss of Signal: Total physical fiber cut or severe macro-bend. Check outside plant with OTDR.
PON (Green)	Blinking / Flashing	Registering: Authentication in progress. If persistent, light power is weak or configuration profile is missing.
PON (Green)	Steady Solid Glow	Connected: Symmetrical

		operational link established successfully.
Power Levels	Below -27 dBm	Power Fault: Dirty connector faces or high-loss splice. Rework connection or clean with a OneClick cleaner.

Table 6.3: ONT Troubleshooting Matrix

CHAPTER 7

FIELD VISIT: BSNL EXCHANGE OFFICE, MANJERI

7.1 Introduction to the Field Visit

A key part of our training was a technical visit to the BSNL exchange office at Manjeri. The visit allowed us to bridge classroom theory with large-scale telecom practices. The visit provided an opportunity to observe how telecommunication services are delivered from the exchange to end users through optical fiber infrastructure.

7.2 Observations and Core Infrastructure

We inspected the active OLT chassis, patch panels, and backup battery systems. The staff explained the process of multiplexing and routing signals into local distribution loops. Behind the OLT, several backend devices are used for network management. The BNG (Broadband Network Gateway) handles user authentication, IP address allocation, routing, and internet session management. The MNGN platform provides carrier-grade packet transport, integrating FTTH broadband, IPTV, and business data links onto a unified network. The OLT delivers internet services to ONTs through optical fiber cables.

7.3 Conclusion of the Visit

In conclusion, the BSNL exchange visit offered indispensable practical insights. Seeing carrier-grade hardware helped us understand the scale of fiber networks and the complexity of modern broadband delivery.

PART CONCLUSION: FIBER OPTIC INFRASTRUCTURE

In summary, this part of the training detailed the physics and engineering behind high-speed optical communications. Beginning with the core waveguiding physics of single-mode fibers, the report systematically evaluated how light pulses are propagated, attenuated, and structured within field-grade multi-core distribution cables. The rigorous analysis of multi-stage passive optical splitters and mathematical power link budgeting demonstrated how a network ensures operational durability while supporting thousands of subscribers. Practical exercises at Bfone Fibernet allowed us to master fusion splicing, color code identification, and OTDR troubleshooting. Furthermore, the field study at the BSNL Exchange Office in Manjeri provided vital structural insight into macro-scale network topology, illustrating the exact handshake between hardware nodes like OLTs, BNG gateways, and consumer-end ONTs. Ultimately, this section showed that reliable gigabit broadband requires both accurate math and high-quality field installation.

Part II

Electrical Home Appliances Maintenance and Troubleshooting

PART INTRODUCTION: DOMESTIC ELECTRICAL ENGINEERING

Electrical appliances form the core operational infrastructure of the modern home. From electrothermal conversion units to complex induction power inverters and single-phase electromagnetic distribution motors, these household systems represent direct physical applications of fundamental thermodynamic and electromagnetic theories. This second module of the report provides a rigorous technical breakdown of consumer appliance electronics, diagnostic logic, and preventive maintenance protocols executed during the industrial training. The goal was to build a solid diagnostic mindset based on electrical theory rather than basic parts-swapping. The chapters ahead map out essential circuit structures (series vs. parallel wiring configurations) and structural protection systems—such as chassis ground earthing, Miniature Circuit Breakers (MCBs), and high-sensitivity Residual Current Circuit Breakers (RCCBs)—designed to mitigate short circuits and hazardous current leakage. We then detail the construction and troubleshooting of seven appliances: the electric iron, induction cooker, mixer grinder, water geyser, ceiling fan, wall fan, and utility pump. Finally, this module presents practical math problems on electricity auditing and electrical charge calculations.

CHAPTER 8 FUNDAMENTALS OF ELECTRICAL SYSTEMS & SAFETY

8.1 Introduction

Modern homes depend on a variety of electrical appliances. Because these systems run daily, they eventually suffer from bearing wear, insulation aging, and wiring faults. This internship was undertaken to bridge the gap between abstract electrical theory and hands-on diagnostic engineering. The training structured under this module focused on developing a core competency in domestic wiring systems, understanding electrical safety protocols, and mastering the systematic teardown, repair, and preventive maintenance of seven foundational household appliances. The primary objective was not merely to fix broken devices, but to understand the physics governing their operation—such as electrothermal conversion, electromagnetism, and fluid dynamics—while strictly adhering to industrial safety standards.

8.2 Fundamentals of Electrical Systems & Safety Overview

A clear understanding of circuit principles and safety gear was established before handling live circuits:

8.2.1 Circuit Behaviors & Laws

- Ohm's Law & Power Equation: Diagnostics rely on Ohm's Law ($V = IR$) and the Power formula ($P = VI = I^2R$). These allow us to calculate expected resistance and identify open or short circuits.
- Series Connections: Current remains constant; voltage divides. Used in safety switches, fuses, and thermostats.
- Parallel Connections: Voltage remains constant; current divides. Used in domestic house wiring so that the failure of one appliance does not shut off others.

$$\mathbf{V = IR \quad \text{and} \quad P = VI = I^2R \quad (8.1)}$$

8.2.2 Earthing and Protection Systems

- Earthing (Grounding): Connects the metallic chassis of an appliance directly to the earth. In case of insulation failure where a live wire touches the metal body, current diverts safely to the ground instead of passing through a user.
- Fuse / MCB (Miniature Circuit Breaker): Protects the circuit from overcurrent and short circuits by breaking the circuit when current exceeds safe thresholds.
- ELCB / RCCB (Residual Current Circuit Breaker): Detects minute current leakages (as low as 30mA) escaping to ground by comparing hot and neutral currents, tripping instantly if an imbalance is found.

CHAPTER 9

TECHNICAL BREAKDOWN & DIAGNOSTICS OF CORE HOME APPLIANCES

9.1 Electric Iron

A. Working Principle

An electric iron utilizes resistive heating (Joule's Effect). Passing current through a high-resistance heating element converts electrical energy into heat ($H = I^2Rt$). Temperature regulation is achieved via a bimetallic strip thermostat. The strip consists of two different metals bonded together with distinct coefficients of thermal expansion. As the iron heats up, one metal expands faster than the other, causing the strip to bend and open the electrical contacts, breaking the circuit. Once it cools down, it straightens, closes the contact, and heat generation resumes.

$$H = I^2Rt \quad (9.1)$$

B. Key Components and Parts

- Soleplate: The heavy, polished aluminum or cast-iron base that transfers heat directly to the clothes.
- Heating Element: A ribbon of Nichrome wound around a mica sheet insulated to prevent electric shock while maximizing heat transfer.
- Bimetallic Thermostat: The temperature control mechanism equipped with an adjustable knob to set specific heat thresholds.
- Thermal Fuse: A backup safety device that permanently melts and opens the circuit if the thermostat fails.

C. Appliance Conclusion

The iron diagnostics clarified resistive heating and bimetallic thermostat feedback. Testing showed the importance of secondary thermal fuses for safety.

9.2 Induction Cooker

A. Working Principle

Induction cooking replaces thermal conduction with electromagnetic induction and localized eddy currents. The appliance takes 50Hz AC power and converts it into a high-frequency (20kHz to 40kHz) alternating current using an internal inverter circuit. This high-frequency current passes through a copper induction coil beneath the ceramic plate, creating a dynamic, rapidly alternating magnetic field. When a ferromagnetic vessel is placed on top, this magnetic field induces circulating eddy currents within the base of the pan. Because the pan has electrical resistance, these currents generate heat directly inside the cookware itself, keeping the stovetop relatively cool.

B. Key Components and Parts

- **Induction Coil:** A flat, spirally wound copper coil that generates the alternating magnetic field.
- **IGBT (Insulated Gate Bipolar Transistor):** The high-speed electronic switch that modulates power and creates the high-frequency current.
- **Main PCB & Microcontroller:** The 'brain' that monitors user inputs, manages power delivery, and reads sensor data.
- **Cooling Fan & RTD (Resistance Temperature Detector):** Critical elements for heat dissipation and monitoring pan temperatures to prevent dry boiling.

C. Appliance Conclusion

Diagnosing induction cookers provided valuable knowledge of high-frequency power electronics. It showed how magnetic fields can be used for clean heating and highlighted PCB-level testing.

9.3 Mixer Grinder

A. Working Principle

The mixer grinder is powered by an AC Universal Motor. This series-wound motor operates on both AC and DC. Current passes through the stator and rotor coils via carbon brushes, creating opposing magnetic fields. This generates the high torque and speeds (up to 20,000 RPM) needed to grind and blend food ingredients.

B. Key Components and Parts

- **Stator & Rotor (Armature):** The fixed and rotating electromagnetic components that generate mechanical torque.
- **Commutator & Carbon Brushes:** The assembly that transfers electrical current from the stationary wires to the rotating armature.

9.3.1 Ceiling Fan

A. Working Principle

The ceiling fan uses a Single-Phase Permanent Split Capacitor Induction Motor. A single-phase source lacks a starting torque, so the motor uses starting and running windings. Connecting a capacitor in series with the starting winding creates a 90-degree phase shift. This simulates a rotating magnetic field, inducing torque in the rotor to rotate the fan.

B. Key Components and Parts

- **Stator & Rotor:** Stationary inner core with starting/running windings and the rotating outer casing.
- **Run/Start Capacitor:** Typically 2.25 μ F to 2.5 μ F capacitor used to split the phase and start rotation.
- **Ball Bearings:** Precision bearings that reduce mechanical friction, ensuring smooth and quiet operation.
- **Down-Rod Assembly:** Structural hardware used to safely suspend the fan from the ceiling.

C. Appliance Conclusion

Ceiling fan repair sessions demonstrated phase separation in single-phase induction. We learned how to isolate electrical winding faults from mechanical bearing seizure.

9.3.2 Wall Fan

A. Working Principle

The wall fan runs on a single-phase capacitor motor, incorporating swiveling oscillation. The motor turns the blades while a gear train or secondary micro-motor sweeps the fan head. Speed adjustment is achieved by selecting different winding taps or through a TRIAC speed controller on the PCB.

B. Key Components and Parts

- Fan Blades and Guard: Aerodynamically pitched plastic or metal blades protected by a wire mesh safety cage.
- Oscillation Gear Assembly / Clutch Knob: A mechanical gear train connected to the rotor shaft that translates high-speed rotation into slow, wide-angle swiveling motion.
- Speed Control Switch (Pull-Cord / Push-Button / Remote PCB): Mechanisms to toggle between different motor speed taps or control electronic voltage regulation.
- Thermal Cut-Off (TCO): A compact thermal fuse embedded directly inside the motor windings to protect the motor from burning out if the blades get obstructed or jammed.

C. Appliance Conclusion

The wall fan provided an study on mechanical linkages combined with induction motors. Diagnostics focused on winding thermal fuses and gear alignment issues.

CHAPTER 10

TROUBLESHOOTING & MAINTENANCE PRACTICE MATRIX

ONT Indicator	Status State	Meaning / Corrective Action
LOS (Red)	Steady Glow	Loss of Signal: Total physical fiber cut or severe macro-bend. Check outside plant with OTDR.
PON (Green)	Blinking / Flashing	Registering: Authentication in progress. If persistent, light power is weak or configuration profile is missing.
PON (Green)	Steady Solid Glow	Connected: Symmetrical operational link established successfully.
Power Levels	Below -27 dBm	Power Fault: Dirty connector faces or high-loss splice. Rework connection or clean with a OneClick cleaner.

Table 6.3: ONT Troubleshooting Matrix

Appliance	Common Symptom	Root Cause Detected	Corrective Action Performed
Electric Iron	No heating; indicator light off	Open circuit in thermal fuse / Broken power cord	Replaced thermal fuse; trimmed and re-terminated power cord.
Induction Cooker	Dead display; won't power up	Blown main fuse due to shorted IGBT	Replaced the shorted IGBT and main glass fuse; checked gate driver circuit.
Mixer Grinder	Sparks from bottom; sudden stop	Worn carbon brushes / Overload switch tripped	Replaced carbon brushes; cleared jar jam and reset the bottom OLP switch.
Water Heater	Water not heating; indicator on	Scale buildup or open-circuit in heating element	Drained tank, removed limescale, and replaced the immersion heating element.
Ceiling Fan	Humming sound; spins very slowly	Degraded or leaked capacitor	Desoldered/ disconnected old capacitor, verified value, and installed

			new 2.5 μ F unit.
Wall Fan	Fan rotates but does not oscillate / Overheating	Stripped linkage gears / Dry rotor shaft	Replaced the plastic gear wheel assembly; cleaned and lubricated the rotor shaft with machine oil.
Pump Motor	Motor hums but does not rotate	Seized bearings / Defective starter capacitor	Freed impeller manually; replaced bearings and capacitor.

Table 10.1: Appliance Troubleshooting & Maintenance Practice Matrix

10.1 Internship Outcomes & Skills Acquired

- **Safety Instinct Diagnostics:** Developed a strict protocol-first mindset—always verifying isolation with a neon tester, utilizing insulation mats, and ensuring appropriate grounding before applying mains voltage.
- **Systematic Fault Profiling:** Shifted away from random guesswork to structured troubleshooting: tracing faults from the source point (plug/cord) through protection systems (OLPs/fuses) to control systems and finally the load.
- **Instrument Competency:** Attained fluid precision in utilizing digital multimeters to measure resistance drops, AC loop voltages, and verifying open/short circuits.

CHAPTER 11

APPLIED PRACTICAL CALCULATIONS

11.1 Domestic Electricity Consumption Auditing

Electricity billing is based on energy consumed in kilowatt-hours (kWh), where 1 Unit represents 1 kWh.

Find the total energy consumed in a 30-day month by a home running two 40W tube lights for 6 hours a day, three 65W ceiling fans for 8 hours a day, and other appliances drawing 150W for 2 hours daily.

$$E_{\text{Lights}} = 2 \times 40 \text{ W} \times 6 \text{ hrs/day} \times 30 \text{ days} = 14,400 \text{ Wh} = 14.4 \text{ kWh}$$

$$E_{\text{Fans}} = 3 \times 65 \text{ W} \times 8 \text{ hrs/day} \times 30 \text{ days} = 46,800 \text{ Wh} = 46.8 \text{ kWh}$$

$$E_{\text{Misc}} = 150 \text{ W} \times 2 \text{ hrs/day} \times 30 \text{ days} = 9,000 \text{ Wh} = 9.0 \text{ kWh}$$

$$E_{\text{Total}} = 14.4 \text{ kWh} + 46.8 \text{ kWh} + 9.0 \text{ kWh} = 70.2 \text{ kWh (Units)}$$

11.2 Charge Calculation in Audio Systems

Determine the total charge transported through a 230V electrical system that consumes 8 MJ of energy during its operating cycle.

$$\text{Given: } V = 230 \text{ V, } E = 8 \times 10^6 \text{ J}$$

$$\text{Equation: } E = qV$$

$$q = E / V$$

$$= 8 \times 10^6 \text{ J} / 230 \text{ V}$$

$$= 3.48 \times 10^4 \text{ Coulombs (C)}$$

PART CONCLUSION: DOMESTIC ELECTRICAL ENGINEERING

To conclude, this second part linked academic physics concepts with practical appliance diagnostics. By systematically mapping out internal component frameworks across seven fundamental consumer appliances, the training provided deep insights into everyday electrical systems. Tracing diagnostic signatures—such as a bimetallic contact opening in a thermal loop or a split-phase capacitor drop in an induction motor—showed how equipment behaviors can be accurately tracked using standard multi-meter analysis. The exercises emphasized the importance of safety devices like thermal cut-offs (TCOs) and overload protectors (OLPs) in preventing damage. Additionally, the quantitative practical math problems confirmed that accurate load auditing and tracking energy changes are key to managing household electricity efficiently. Ultimately, the hands-on repair skills and diagnostic logic learned during this course form a valuable toolkit for electrical maintenance.